

Table II: Microcontroller Series from Some Other Manufacturers

NXP Semiconductors			Renesas Electronics				STMicroelectronics			Texas Instruments			Dallas Semiconductor	
8 Bit	16 Bit	32 Bit	4 Bit	8 Bit	16 Bit	32 Bit	8 Bit	16 Bit	32 Bit	8 Bit	16 Bit	32 Bit	8 Bit	16 Bit
LPC700, LPC900 Series	XA Series	ARM7: LPC2100, LPC2200, LPC2300, LPC2400 Series	720 Family	78K Family	RL78 Family	RX Family	ST6 Family	ST10 Family	ST20 Family	TMS370 Family	MSP430 Family	TMS320 (DSP) Family	8051 Family	MAXQ RISC Family
		ARM9: LPC2900, LPC3100, LPC3200 Series		H8/SLP Family		M16C Family	SuperH Family	ST7 Family	ARM7: STR7 (ARM7TDMI)		C2000 Family			
		ARM Cortex-M0: LPC1100, LPC1200 Series		740 Family		H8 Family	V850 Family	STM8 Family	ARM9: STR9 (ARM966E-S)		Stellaris (ARM Cortex-M3)			
		ARM Cortex-M0+ : LPC800 Series				R8C Family	R32C Family	μPSD (8032) Family	ARM Cortex-M0: STM32 F0		Tiva C Series			
		ARM Cortex-M3: LPC1300, LPC1700, LPC1800 Series		78K0R Family		H8SX Family			ARM Cortex-M3: STM32 F1, F2, L1, W		Hercules - TMS570 (ARM Cortex-R4), TMS470M ARM Cortex-M3, RM4 ARM Cortex-R4			
		ARM Cortex-M4: LPC4000, LPC4300 Series							ARM Cortex-M4: STM32 F4					

SoC. PandaBoard ES is a newer version based on OMAP4460 SoC, with its CPU and GPU running at higher clock rates.

OMAP4430 SoC on PandaBoard features a dual-core 1GHz ARM Cortex-A9 MPCore CPU, a 304MHz PowerVR SGX540 GPU, IVA3 multimedia hardware accelerator with a programmable DSP, and 1GB DDR2 SDRAM.

PandaBoard ES uses a newer SoC, with a dual-core 1.2GHz CPU and 384MHz GPU. Primary persistent storage is via an SD card slot, which supports SDHC cards up to 32 GB. The board includes wired 10/100 Ethernet as well as wireless Ethernet and Bluetooth connectivity.

The device runs the Linux kernel,

with either traditional distributions, Android or Mozilla Firefox OS. Optimised versions of Android and Ubuntu are available from the Linaro Foundation. Linaro has selected PandaBoard as one of the hardware platforms that it supports with monthly build images.

BeagleBoard. BeagleBoard is a low-power, open-source hardware, single-board computer produced by Texas Instruments in association with DigiKey and Newark element14. It was also designed with open source software development in mind, and as a way of demonstrating Texas Instruments' OMAP3530 SoC.

OMAP3530 includes an ARM Cortex-A8 CPU, a TMS320C64x+ DSP for accelerated video and audio decoding, and an Imagination Technologies' PowerVR SGX530 GPU to provide accelerated 2D and 3D rendering that supports OpenGL ES 2.0. Video-out is provided through separate S-Video and HDMI connections.

A single SD/MMC card slot supporting SDIO, a USB On-The-Go port, an RS232 serial connection, a JTAG connection, and two stereo 3.5 mm jacks for audio in/out are also provided.



Fig. 7: BeagleBoard

Atmel STK600. It is a complete starter kit and development system for 8-bit and 32-bit AVR microcontrollers that gives designers a quick start to develop code on the AVR with advanced features for prototyping and testing new designs. The AVR device connects to STK600 using an innovative routing and socket card sandwich system, which routes signals from the device to the appropriate hardware.

The system consists of a generic socket card, on which the AVR device is inserted, and a device-specific signal routing card, which routes signals from socket pins to different functions



Fig. 6: PandaBoard